

COVER PAGE

IFB - Interface Board

Schematic

Version: 1.0.2



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: CoverPage.SchDoc	Sheet 1 of 34
Checked by: Bao Bui	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026

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NOTE:

/DNP: Do Not Place

- V1.0.0

- V1.1.0: Add FPGA - Change to SAMA7G54



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: Index.SchDoc

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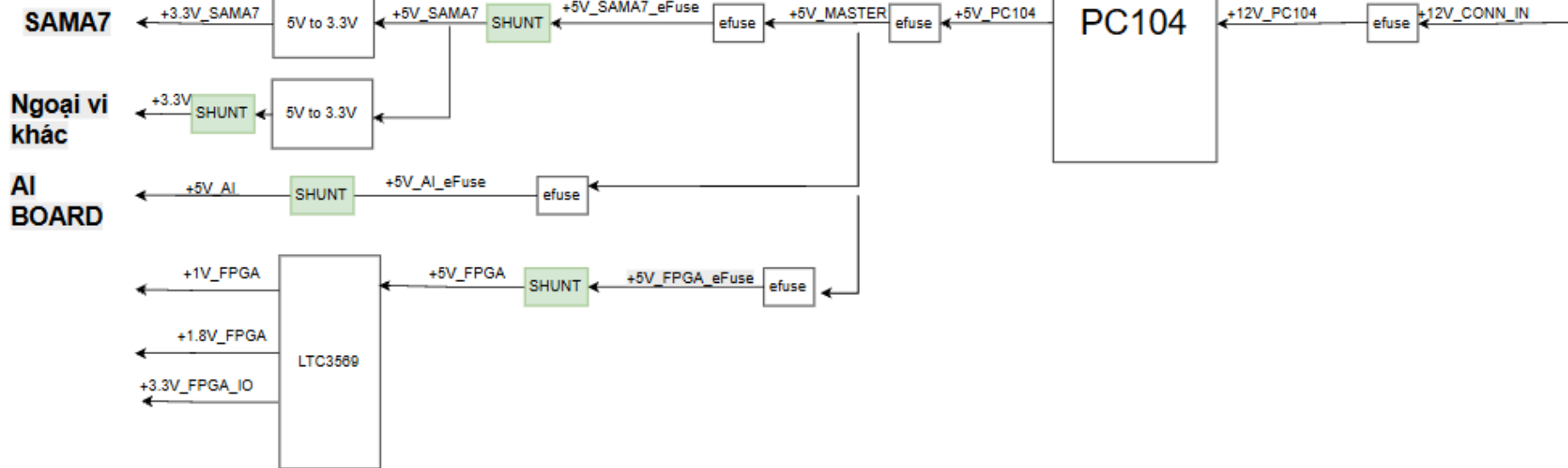
Document No: 2

Size: A3

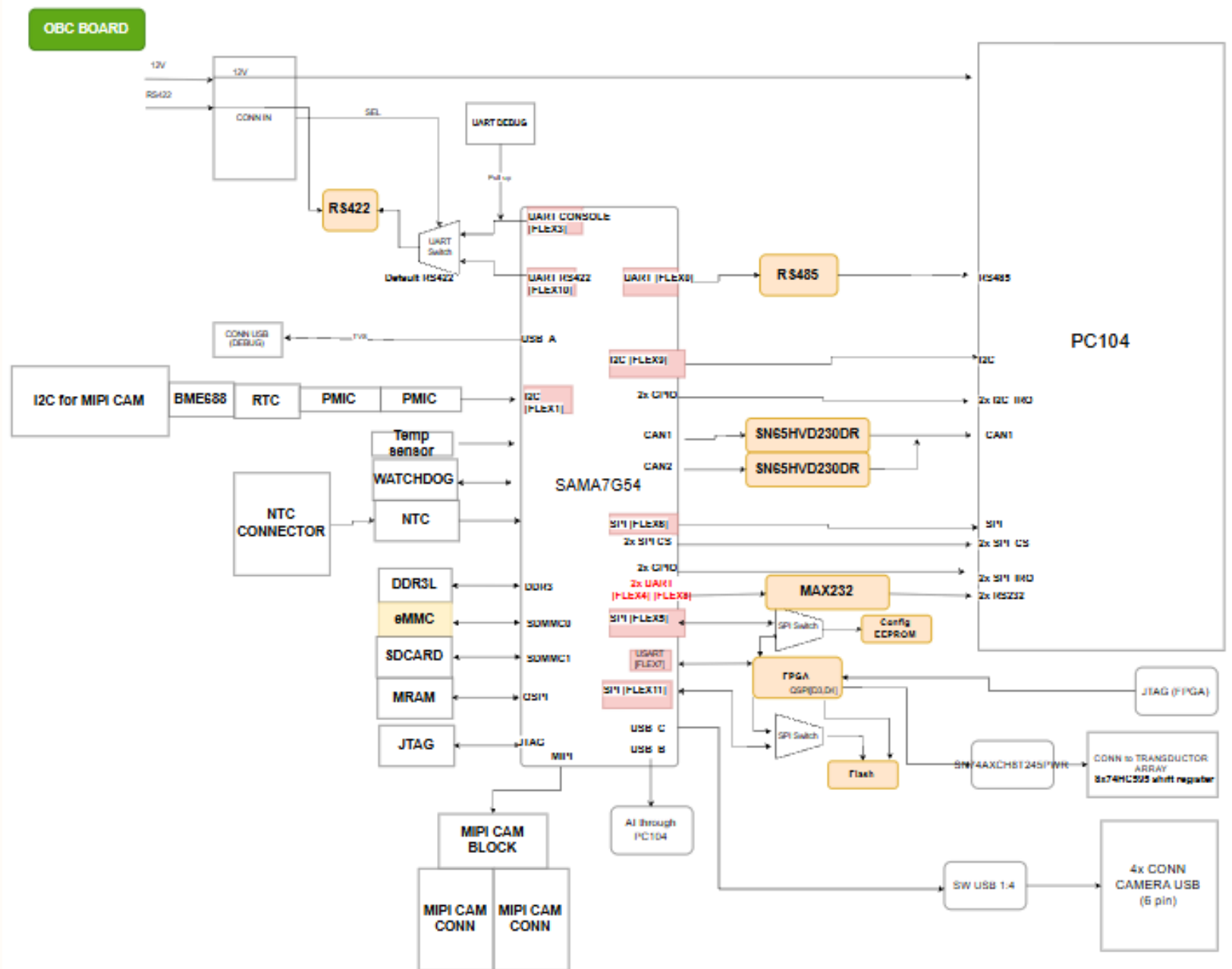
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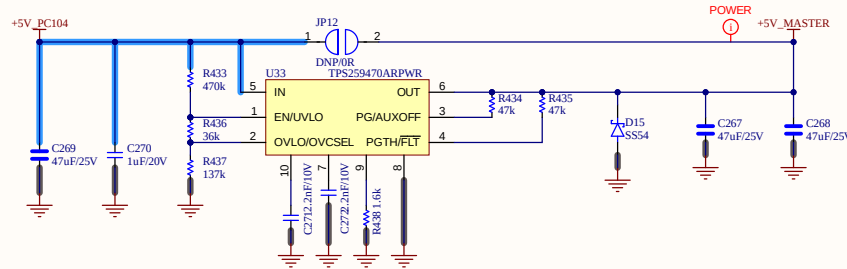
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: PowerTree.SchDoc	Sheet 3 of 34
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Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026



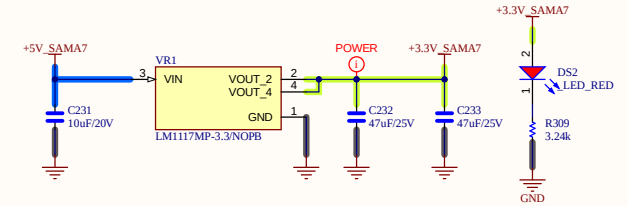
2i-Engineering/View-PCRs. This repository contains the Altium source for View's PCBs and adapter board Q1V0-CB-A, 2iV0-CB-B, 3iV0-CB-C, and 2iV0-CB-D.

EFUSE for 5V MASTER

Undervoltage Protector: NONE
Overvoltage Protector: 1.2*(133+470)/133=5.44V
Slew Rate: 2000/2200=0.91V/ms
I-LIM: 3334/1600=2.08A

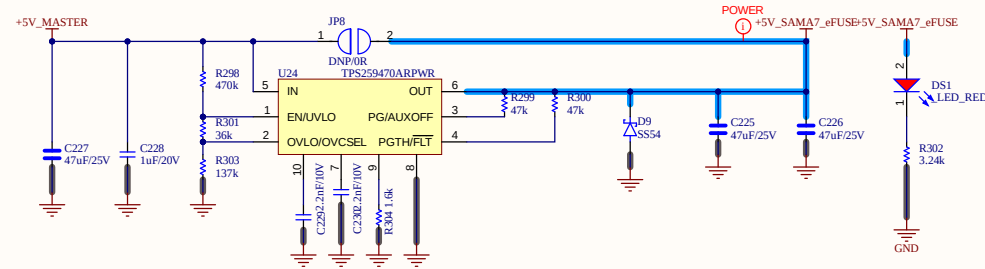


LD0 for SAMA7

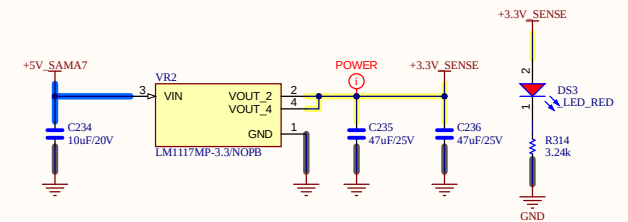


EFUSE for SAMA7

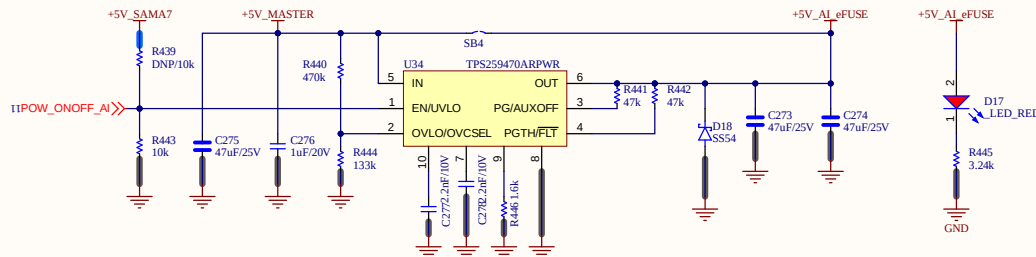
Undervoltage Protector: NONE
Overvoltage Protector: 1.2*(133+470)/133=5.44V
Slew Rate: 2000/2200=0.91V/ms
I-LIM: 3334/1600=2.08A



LD0 3.3V

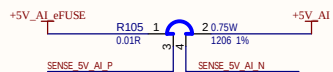
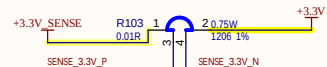
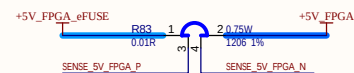
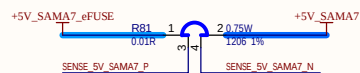
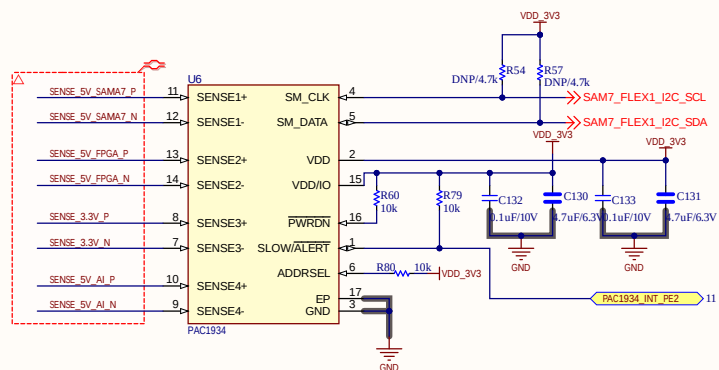


EFUSE for 5V AI



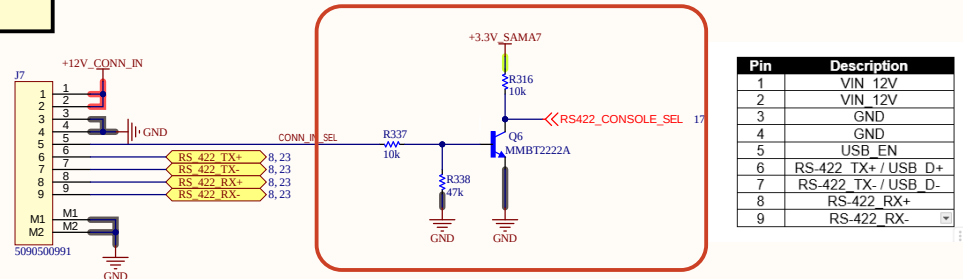
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: Power.SchDoc	Sheet 5 of 34
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POWER MEASUREMENT 12V AND 5V FPGA

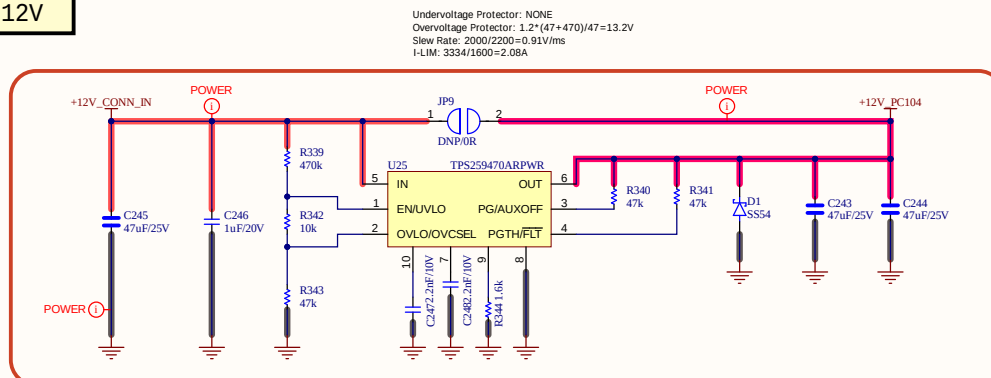


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: POWER_MANAGER.SchDoc	Sheet 6 of 34
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CONN IN



EFUSE 12V



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: CONNECTOR_IN.SchDoc Sheet 7 of 34

Checked by: Bao Bui

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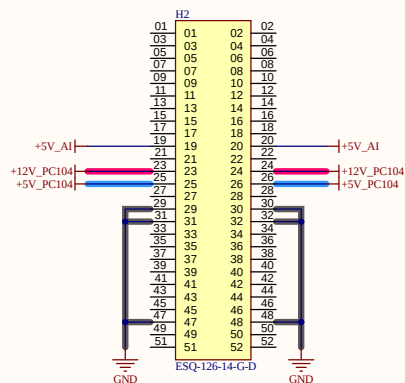
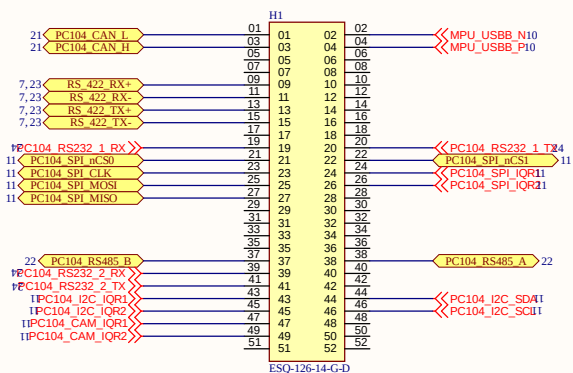
Size: A3

Approved by: Vu Pham

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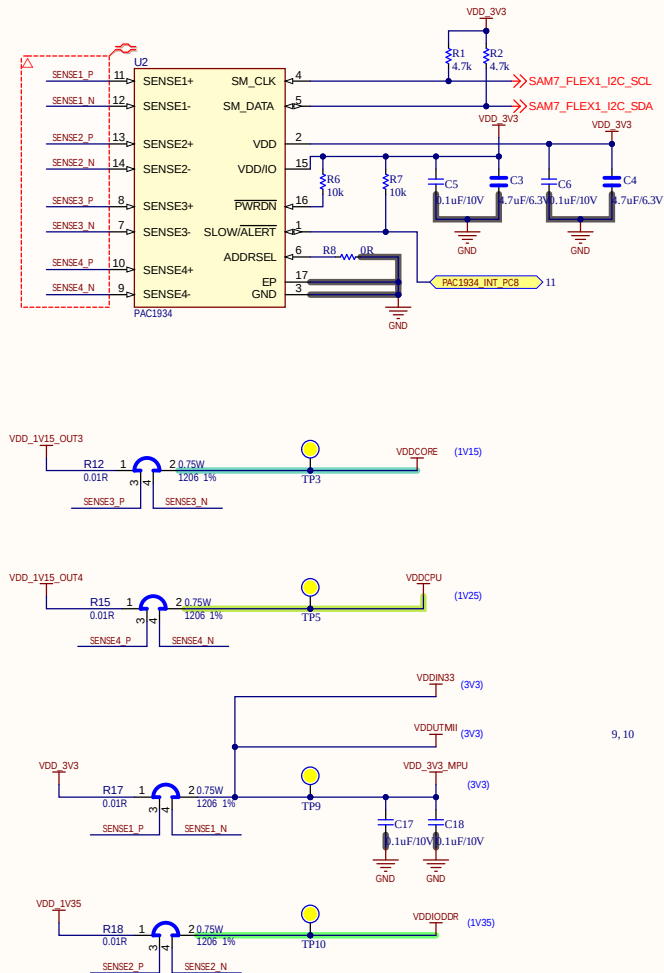
Date: 07/07/2026

PC104

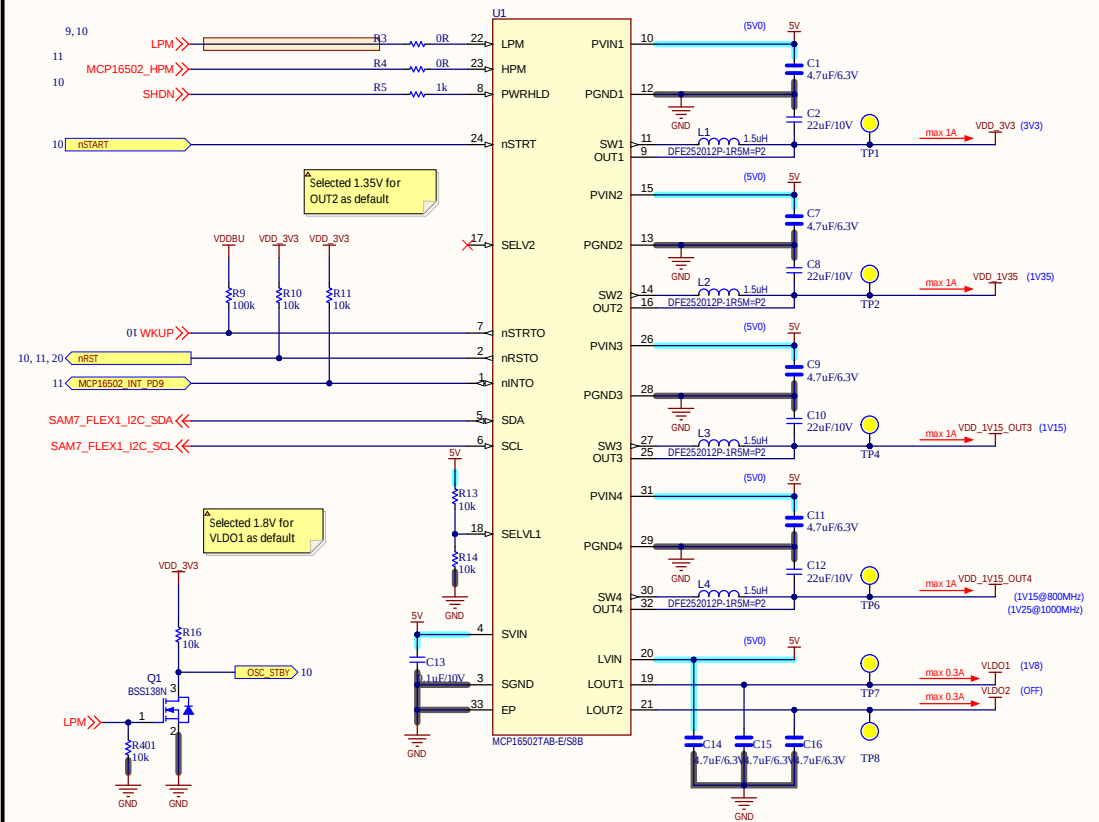


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Design by: Cam Doan	Sheet title: CONNECTOR_BUS.SchDoc	Sheet 8 of 34
Checked by: Bao Bui	Document No: 8	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026

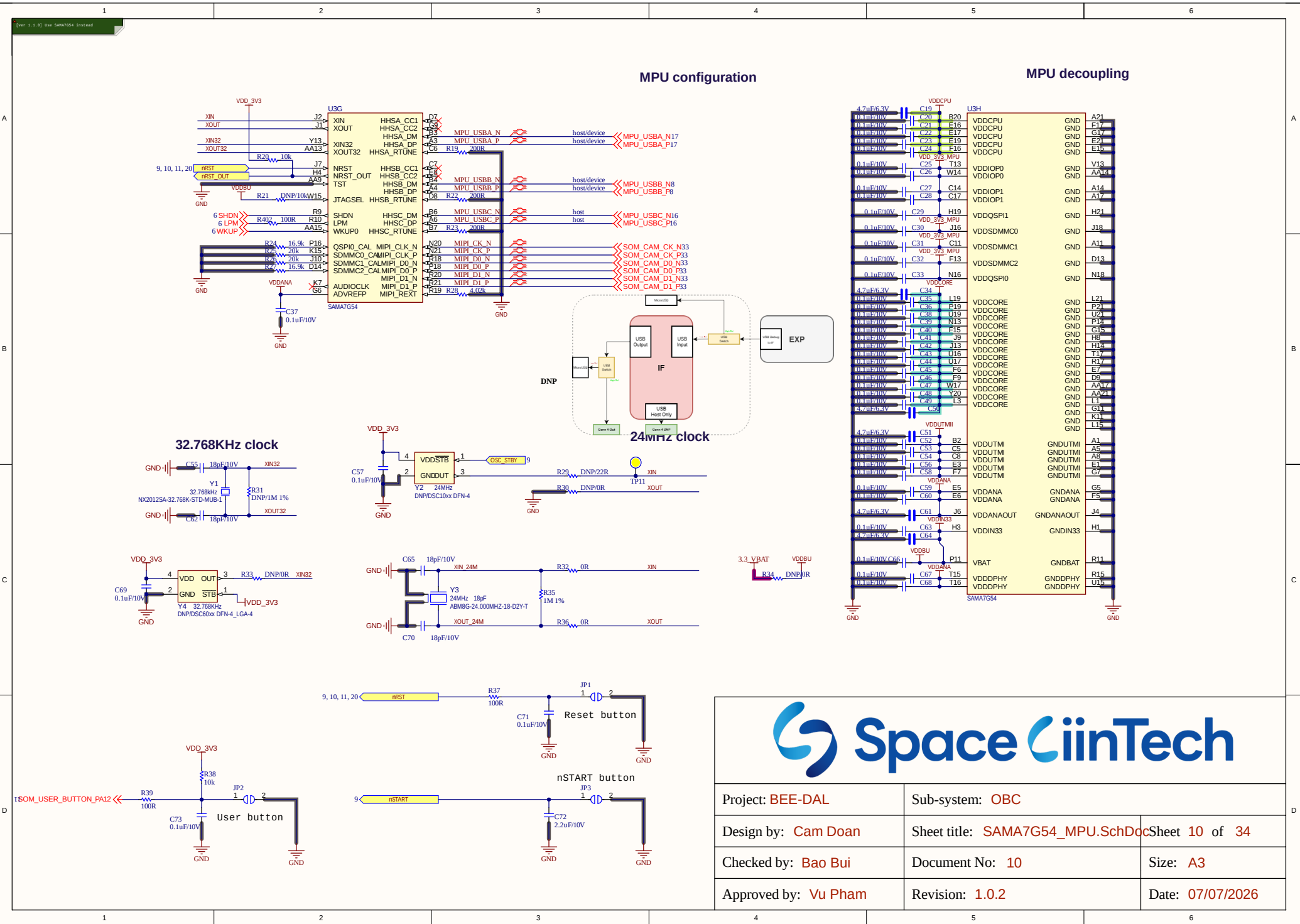
Power Measurement and Distribution



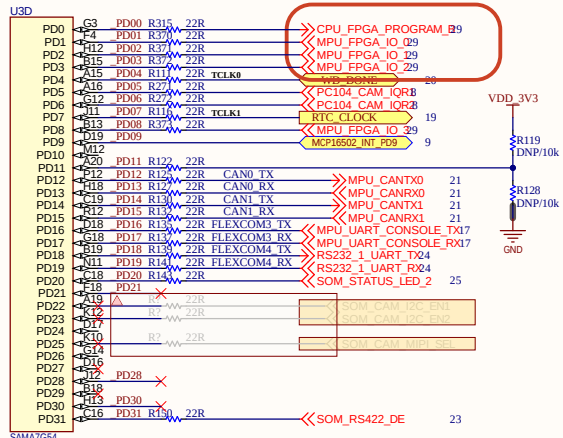
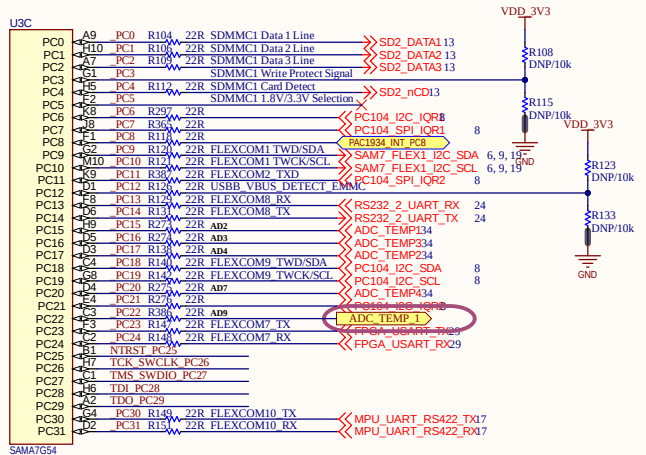
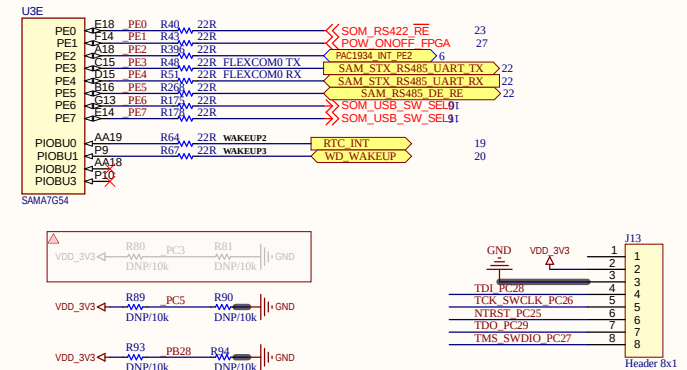
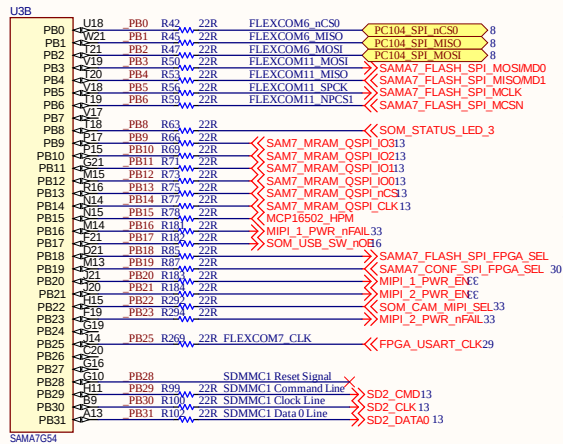
PMIC



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Design by: Cam Doan	Sheet title: SAMA7G54_PMIC.SchDoc	Sheet 9 of 34
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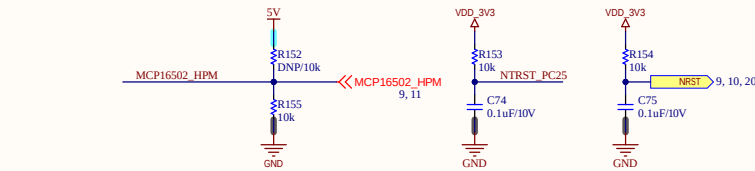
SAMA7G5 Bus Distribution



SPI:4 F5: SPI SW EEPROM
F6: PC104
F11: SPI SW FLASH

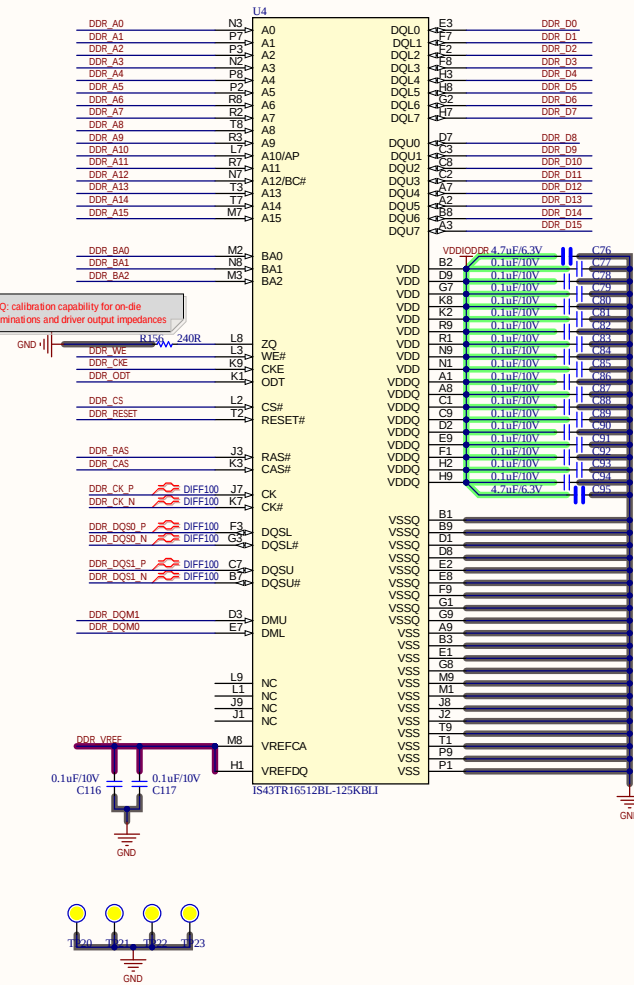
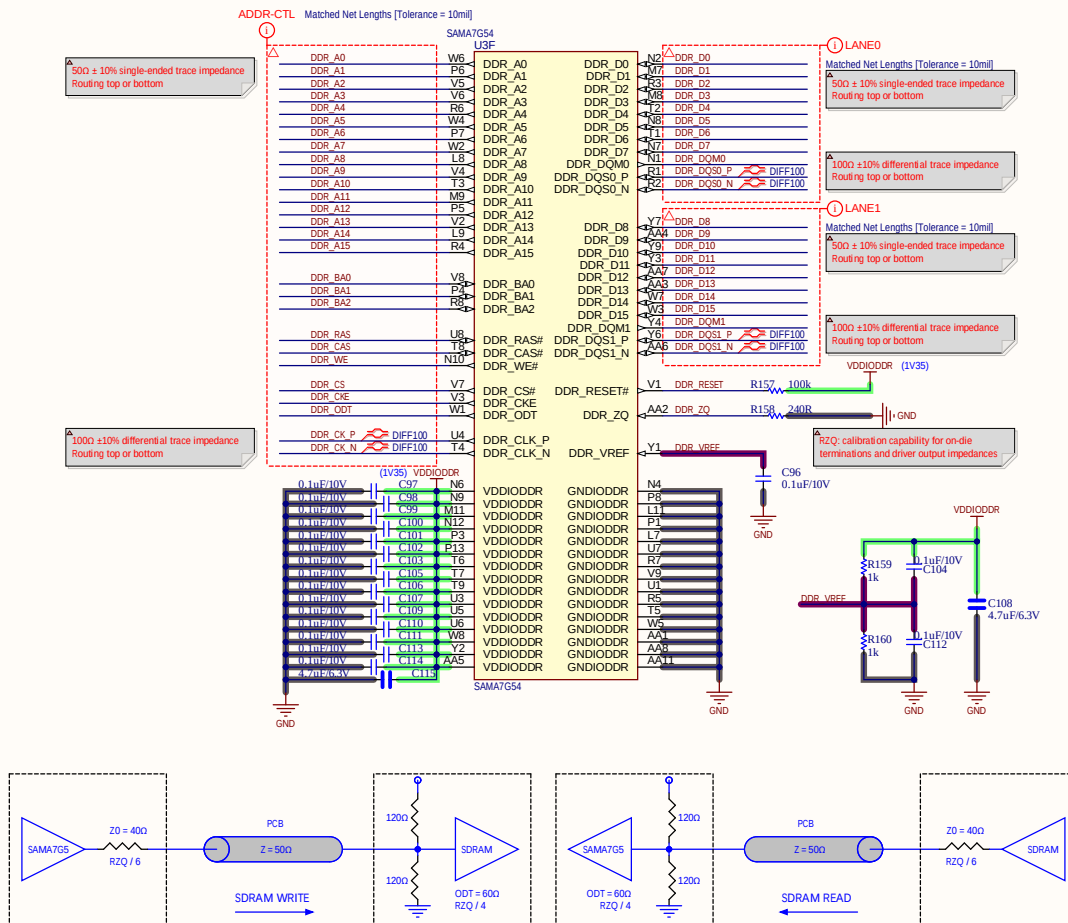
UART:6 F0: RS485
F3: CONSOLE
F10: RS422
F4: PC104
F8: PC104
F7: FPGA

I2C:2 F1: RTC+2xPMIC
F9: PC104



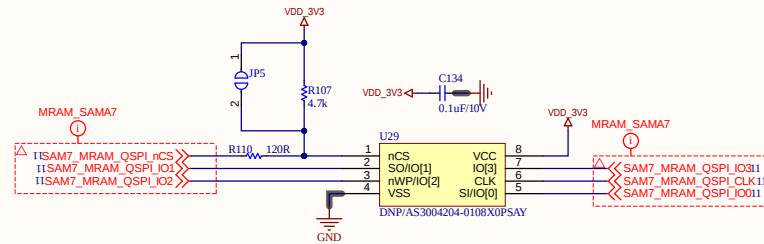
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: SAMA7G54_PIO.SchDoc	Sheet 11 of 34
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SAMA7G5 DDR3-1066 controller

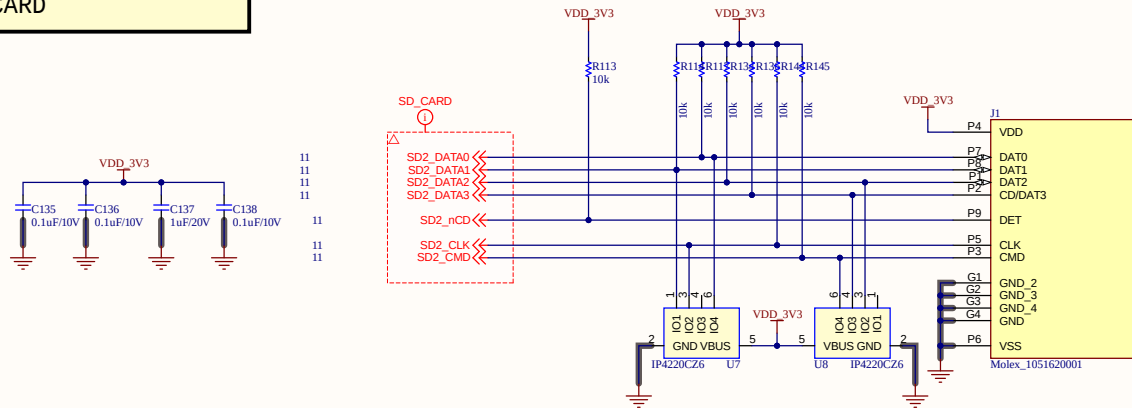


Project: BEE-DAL	Sub-system: OBC
Design by: Cam Doan	Sheet title: SAMA7G54_DDR3L_Sch Sheet 12 of 34
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	Date: 07/07/2026

MRAM 4Mbit



SD CARD



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: SDCARD_MRAM.SchDoc Sheet 13 of 34

Checked by: Bao Bui

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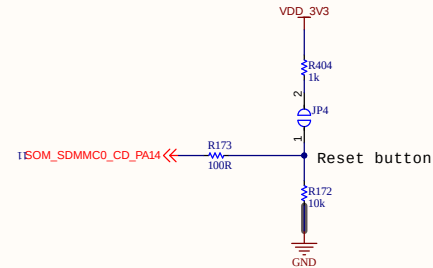
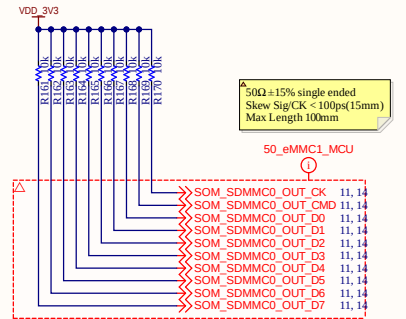
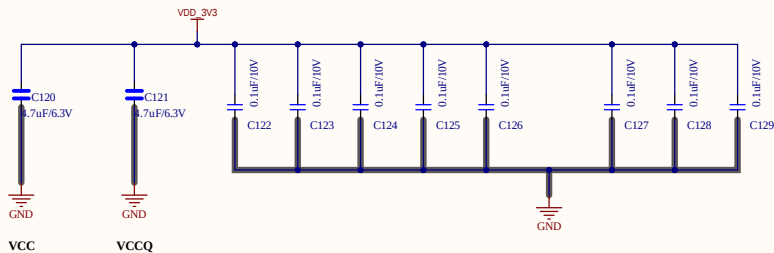
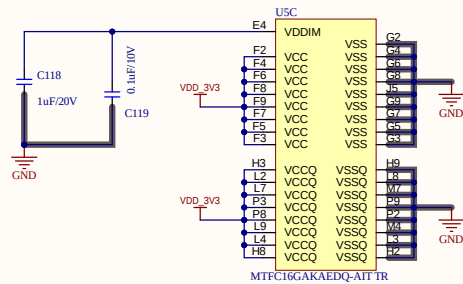
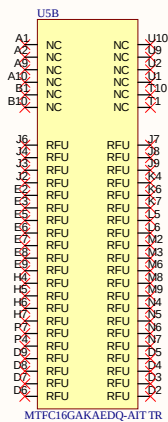
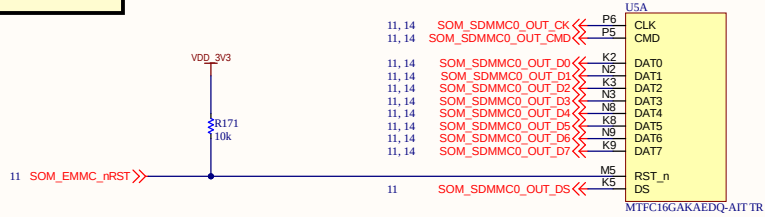
Size: A3

Approved by: Vu Pham

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eMMC - Buffer



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Design by: Cam Doan	Sheet title: eMMC.SchDoc	Sheet 14 of 34
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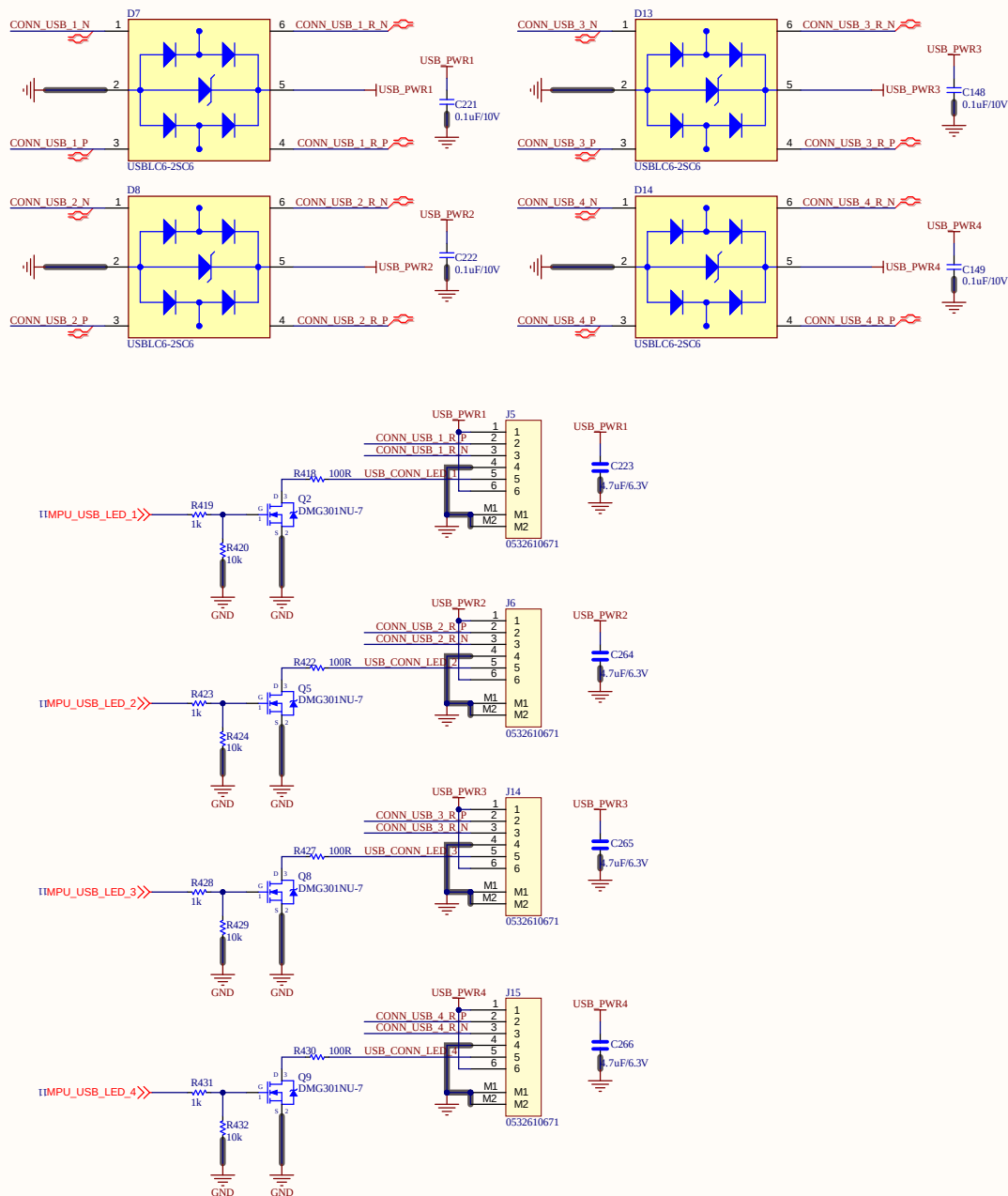
1	2	3	4	5	6	
A						A
B						B
C						C
D						D
1	2	3	4	5	6	

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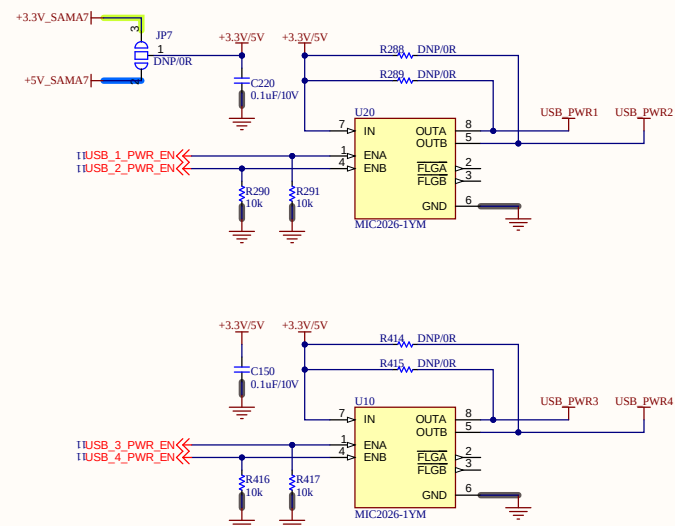


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Design by: Cam Doan	Sheet title: USB_CAMERA_2.SchDoc	Sheet 15 of 34
Checked by: Bao Bui	Document No: 15	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026

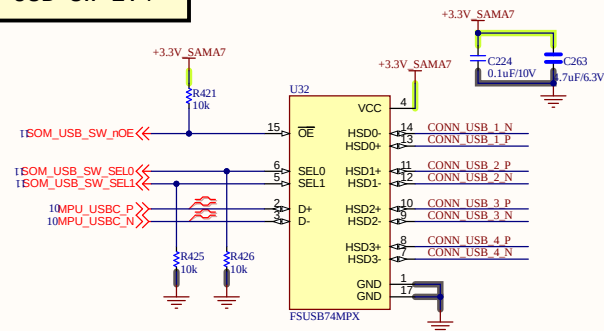
USB Camera



POWER SWITCH



USB SW 1:4



dùng N mos đk
2 mipi cam
3 usb cam
usb len AI board
eFuse cho 5V board AI t



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: USB_CAMERA.SchDoc	Sheet 16 of 34
Checked by: Bao Bui	Document No: 16	Size: A3
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UART MUX 4:1

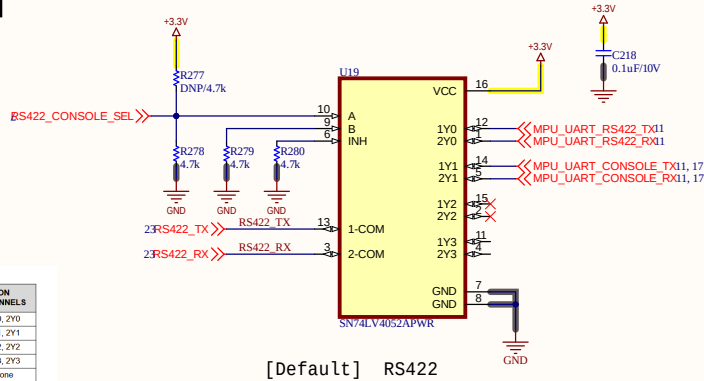
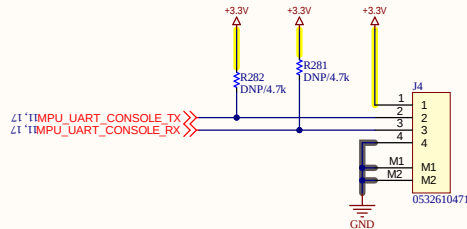


Table 7-1. Function Table

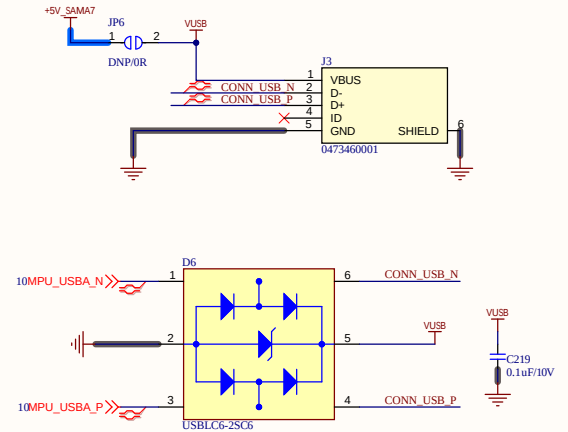
INPUTS			ON CHANNELS
INH	B	A	
L	L	L	1Y0, 2Y0
L	L	H	1Y1, 2Y1
L	H	L	1Y2, 2Y2
L	H	H	1Y3, 2Y3
H	X	X	None

[Default] RS422

UART CONSOLE

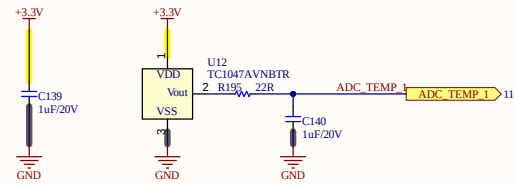


USB CONSOLE

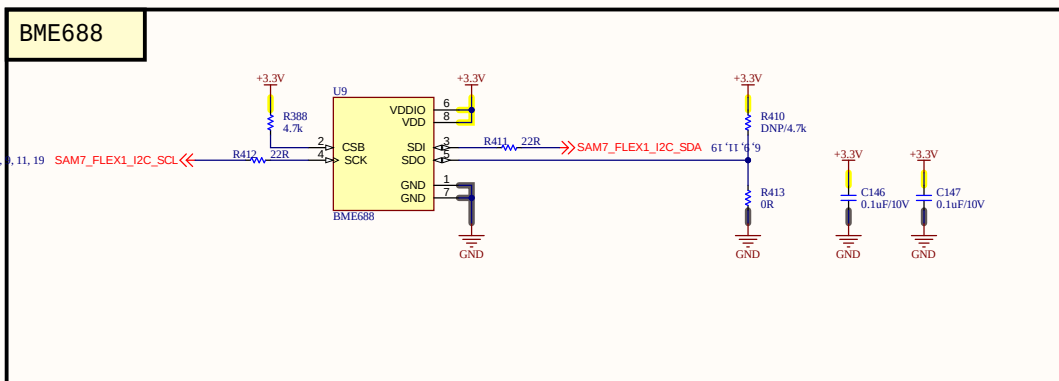
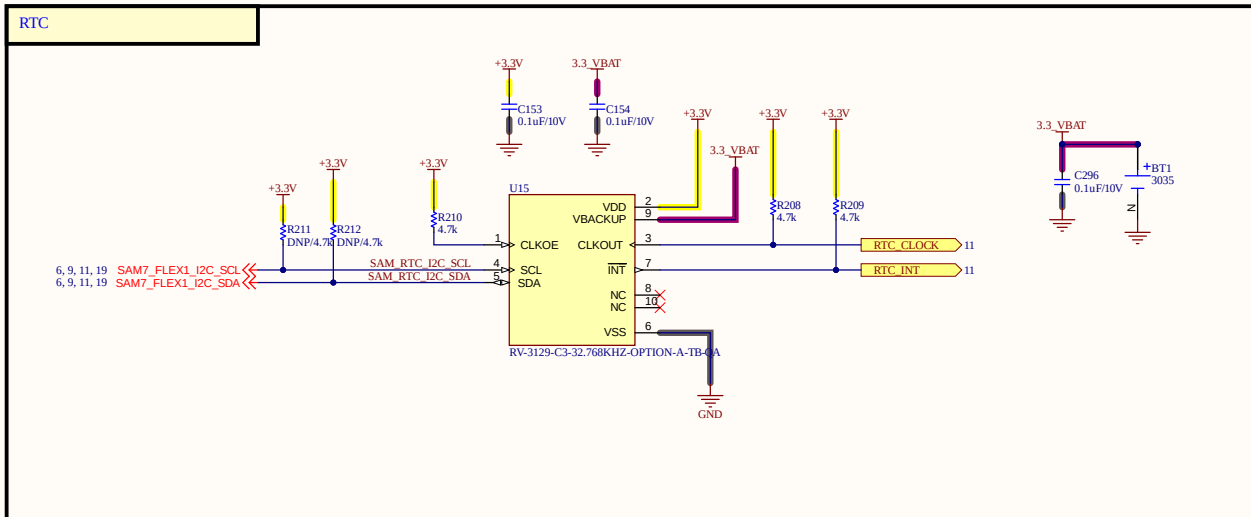


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Design by: Cam Doan	Sheet title: DEBUG_CONSOLE.SchDoc	Sheet 17 of 34
Checked by: Bao Bui	Document No: 17	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026

SENSOR TEMP

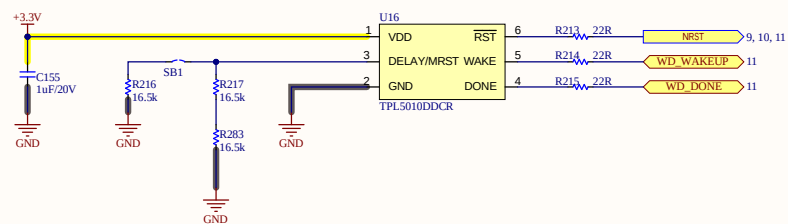


Project: BEE-DAL	Sub-system: OBC	
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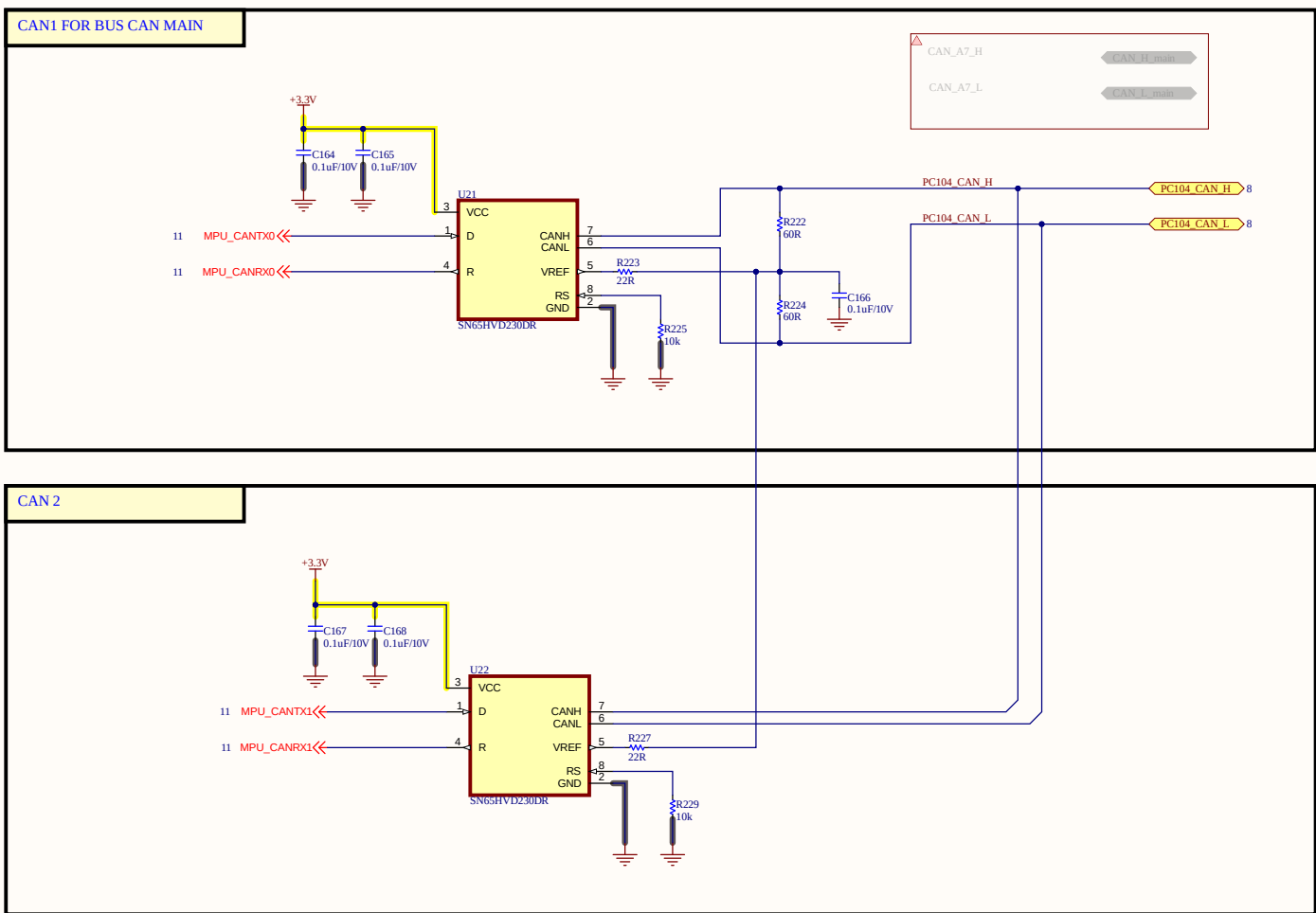


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: RTC.SchDoc	Sheet 19 of 34
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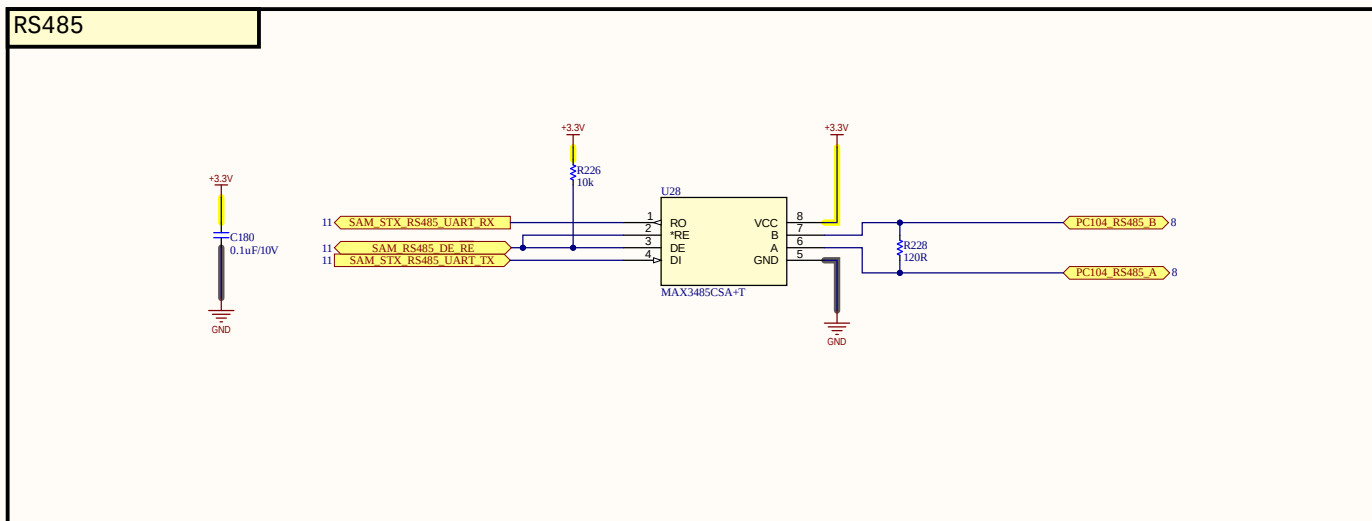
WATCHDOG



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Design by: Cam Doan	Sheet title: WATCHDOG.SchDoc	Sheet 20 of 34
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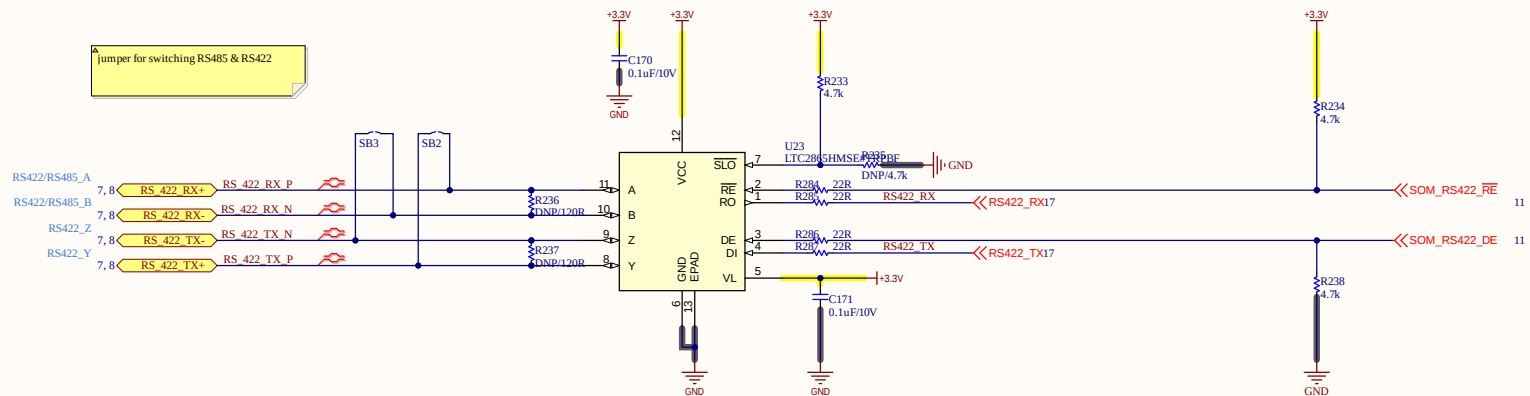
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: CAN_PHY.SchDoc	Sheet 21 of 34
Checked by: Bao Bui	Document No: 21	Size: A3
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Design by: Cam Doan	Sheet title: RS485.SchDoc	Sheet 22 of 34
Checked by: Bao Bui	Document No: 22	Size: A3
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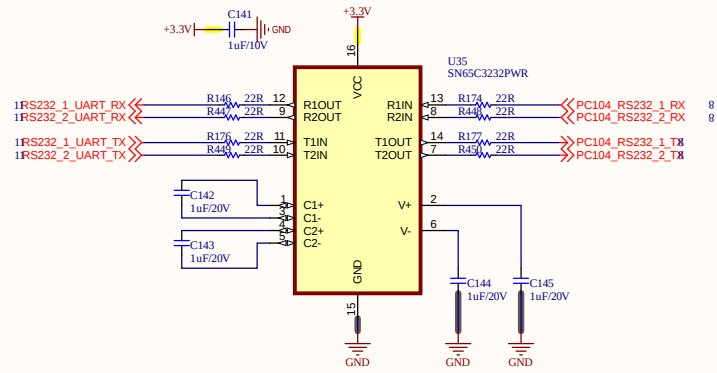
RS422

Jumper for switching RS485 & RS422

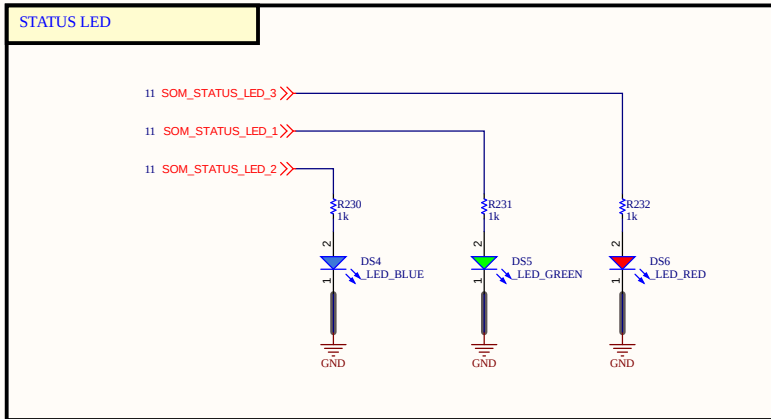


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: RS422.SchDoc	Sheet 23 of 34
Checked by: Bao Bui	Document No: 23	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026

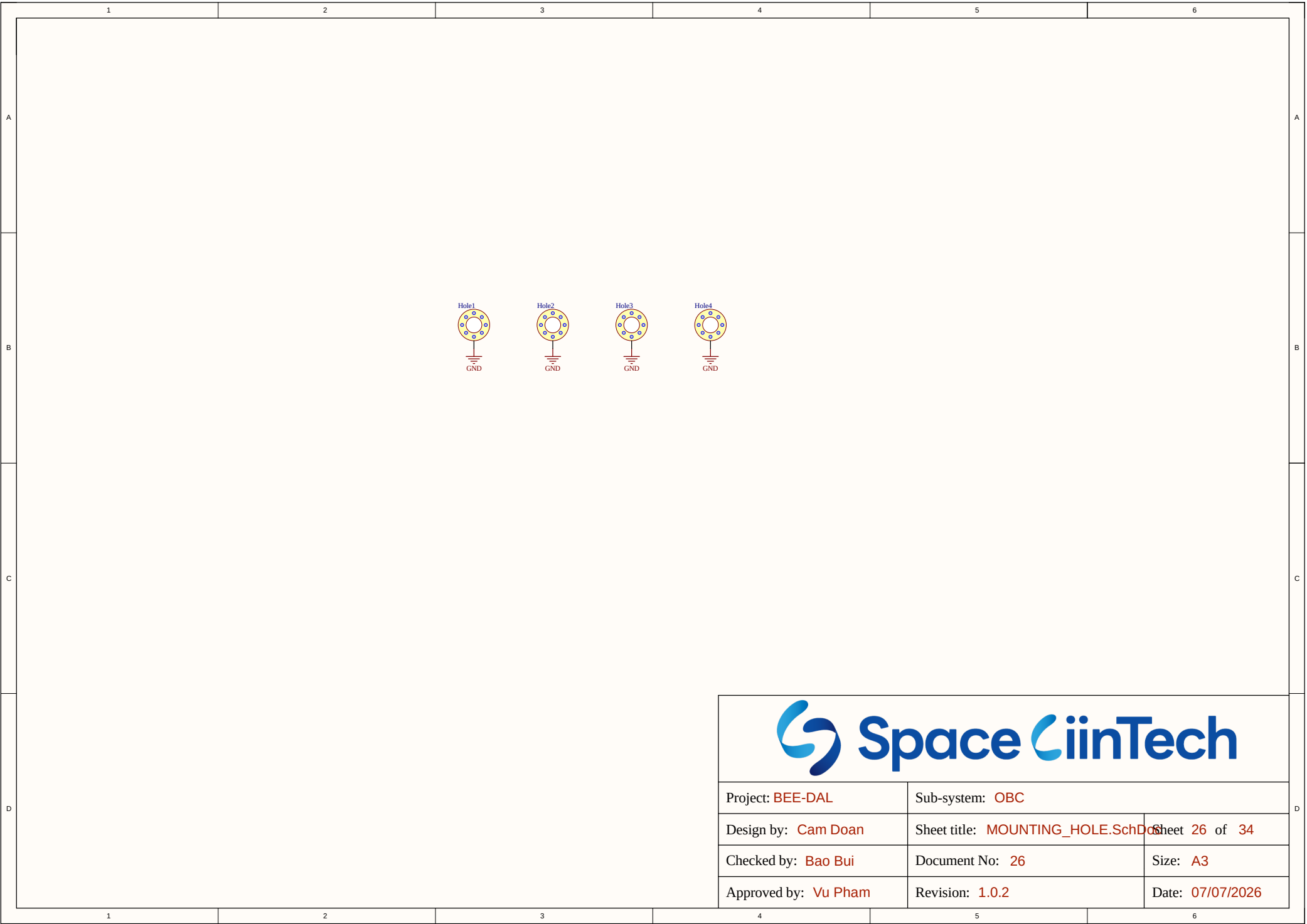
RS232



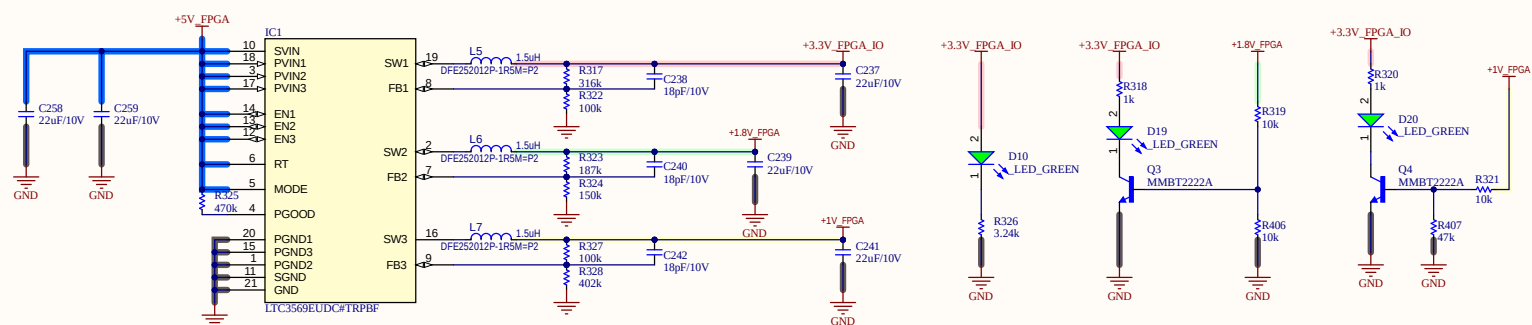
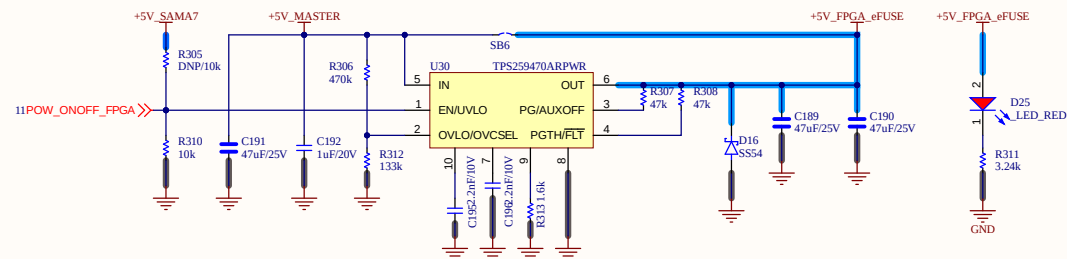
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: RS232.SchDoc	Sheet 24 of 34
Checked by: Bao Bui	Document No: 24	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026



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Design by: Cam Doan	Sheet title: BUTTON_LED.SchDoc	Sheet 25 of 34
Checked by: Bao Bui	Document No: 25	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026



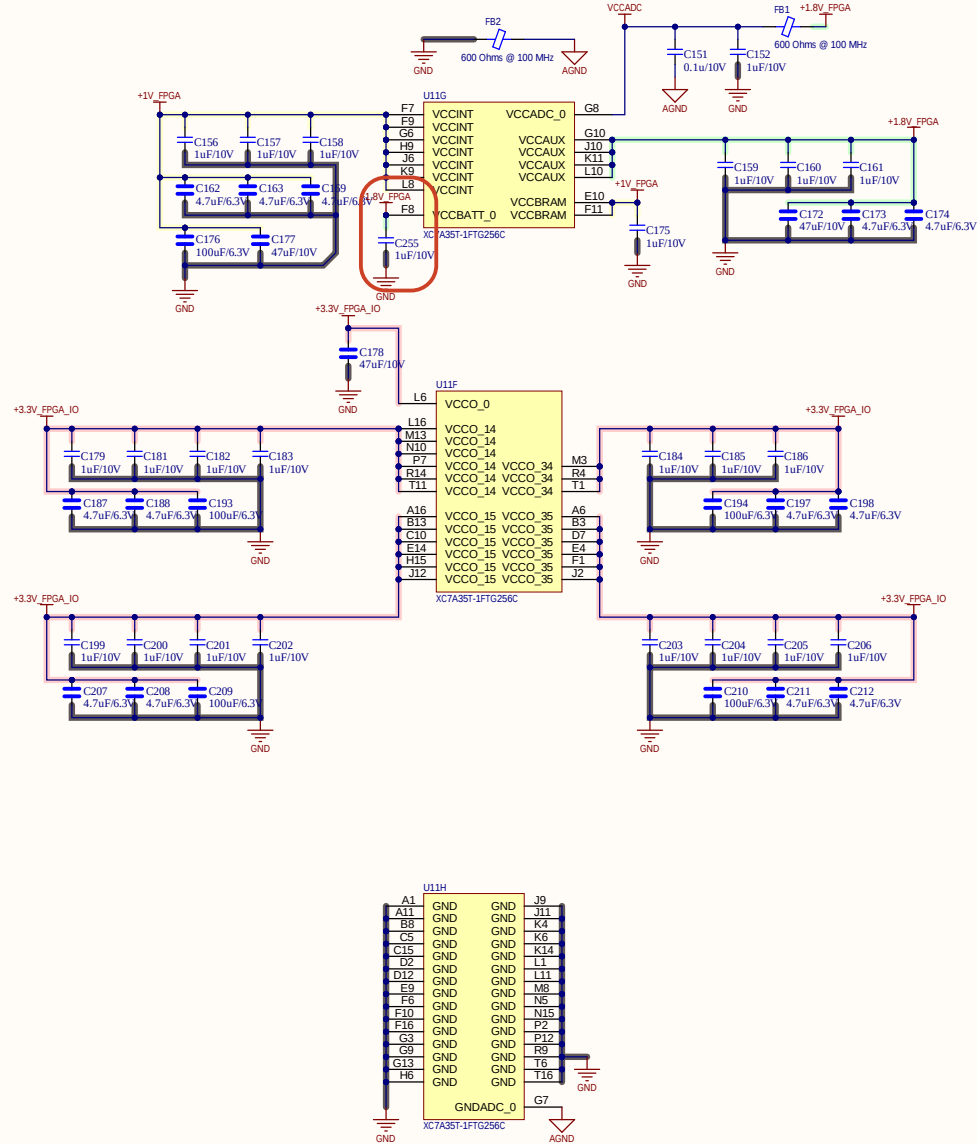
Main Source 5V



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: FPGA_POWER.SchDoc	Sheet 27 of 34
Checked by: Bao Bui	Document No: 27	Size: A3
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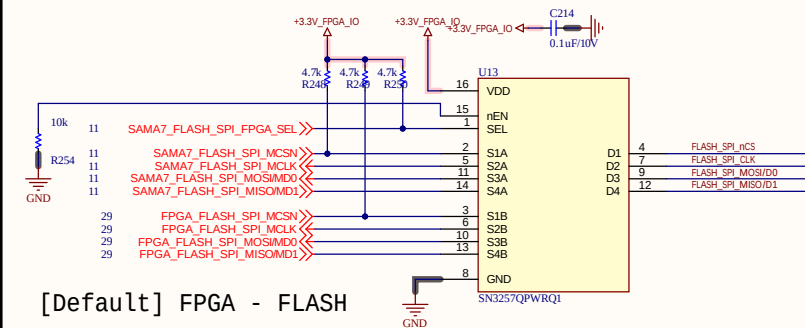
Table 3.1. VCCIO Supply Options

VCCIO Bank	Selection Jumper	3.3 V	1.8 V	1.5 V	1.2 V
VCCIO0	J47	Selectable	Default	—	—
VCCIO1	—	Fixed	—	—	—
VCCIO2	—	Fixed	—	—	—

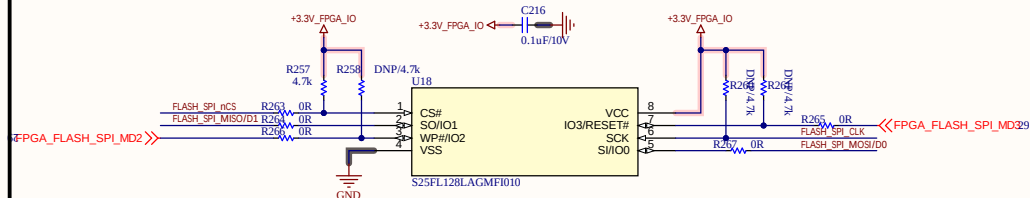


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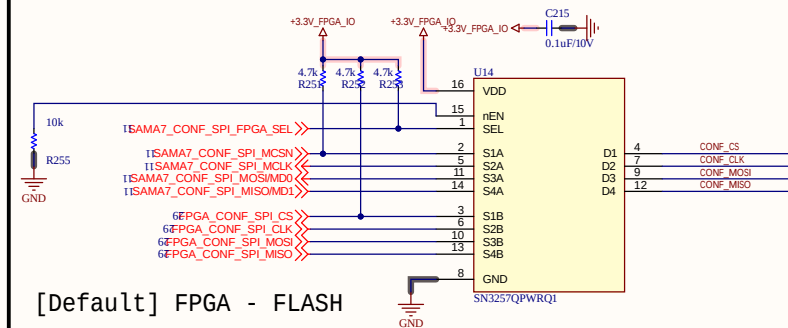
FPGA FLASH



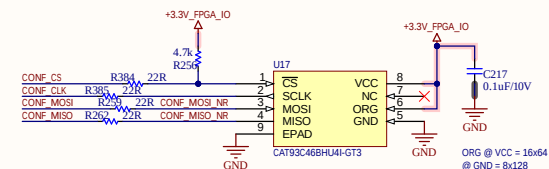
[Default] FPGA - FLASH



FPGA EEPROM



[Default] FPGA - FLASH



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: FPGA_FLASH_EEPROM, Sheet 080 of 34

Checked by: Bao Bui

Document No: 30

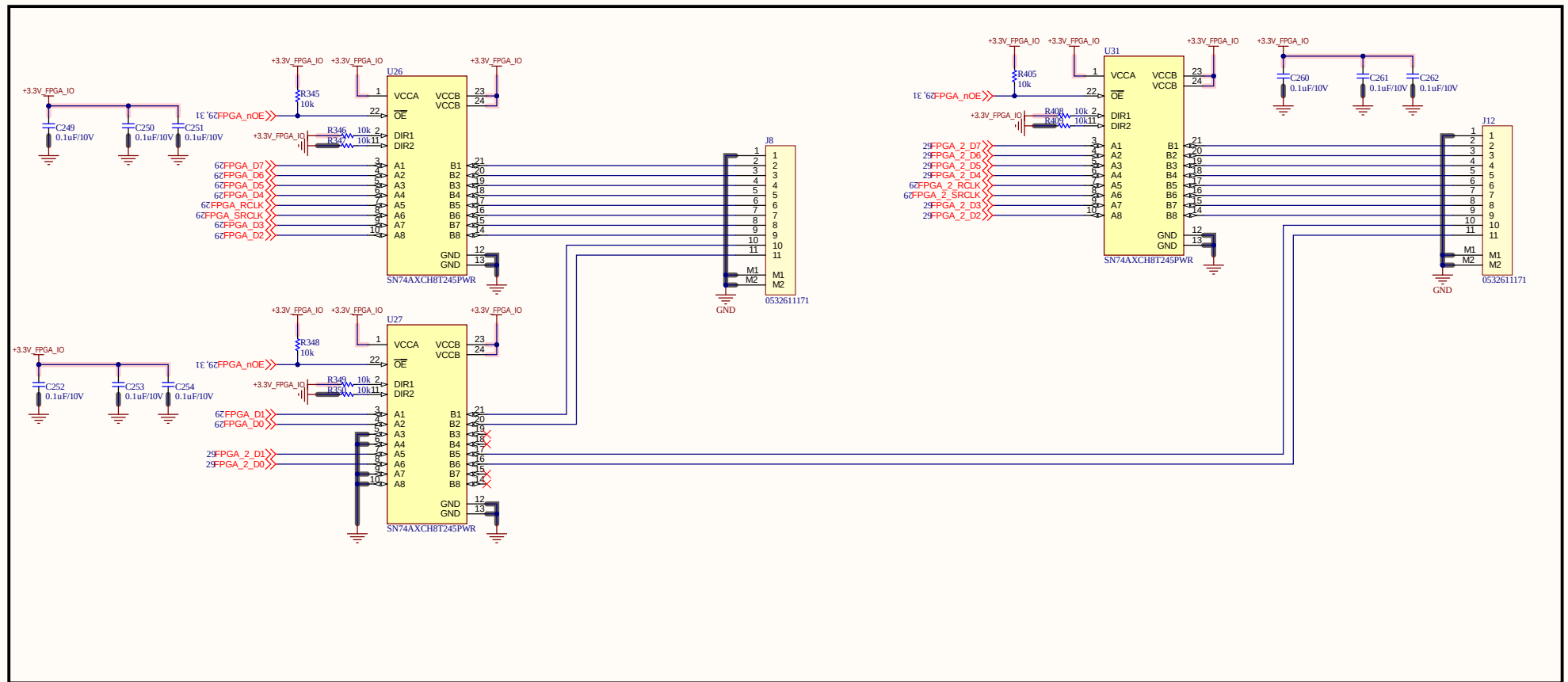
Size: A3

Approved by: Vu Pham

Revision: 1.0.2

Date: 07/07/2026

L	L	H	B to A	A to B
L	H	L	A to B	B to A



Project: BEE-DAL	Sub-system: OBC
Design by: Cam Doan	Sheet title: FPGA_CONN_OUT.SchDoc Sheet 31 of 34
Checked by: Bao Bui	Document No: 31
Approved by: Vu Pham	Revision: 1.0.2
	Size: A3
	Date: 07/07/2026

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: ChangeNote.SchDoc

Sheet 32 of 34

Checked by: Bao Bui

Document No: 32

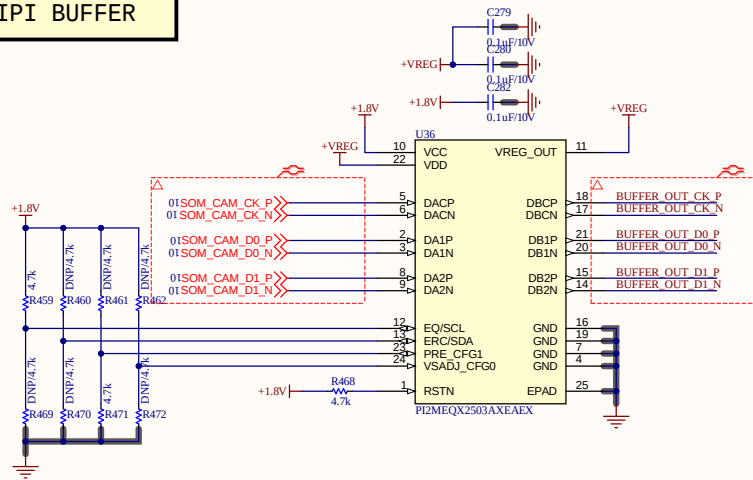
Size: A3

Approved by: Vu Pham

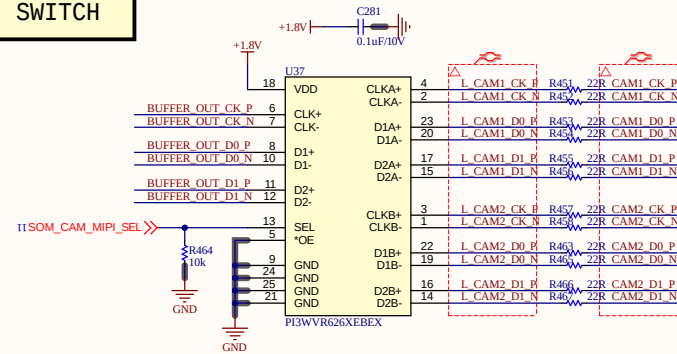
Revision: 1.0.2

Date: 07/07/2026

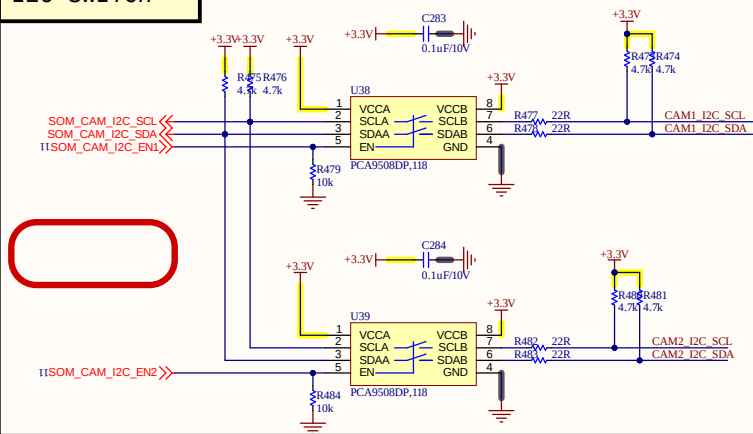
MIPI BUFFER



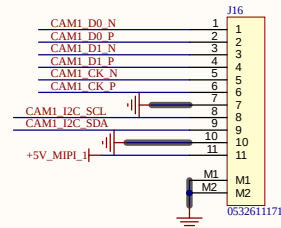
MIPI SWITCH



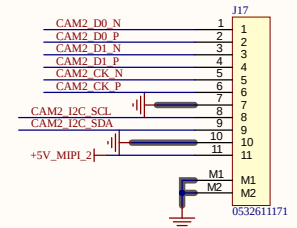
I2C SWITCH



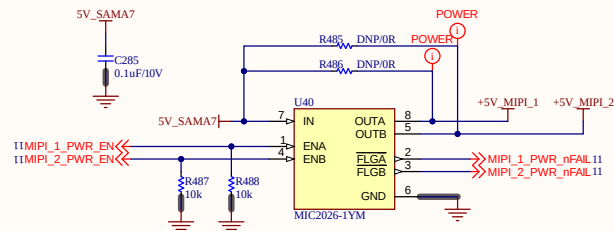
for CAMERA 1



for CAMERA 2



POWER SWITCH



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: MIPI_CSI.SchDoc

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Checked by: Bao Bui

Document No: 33

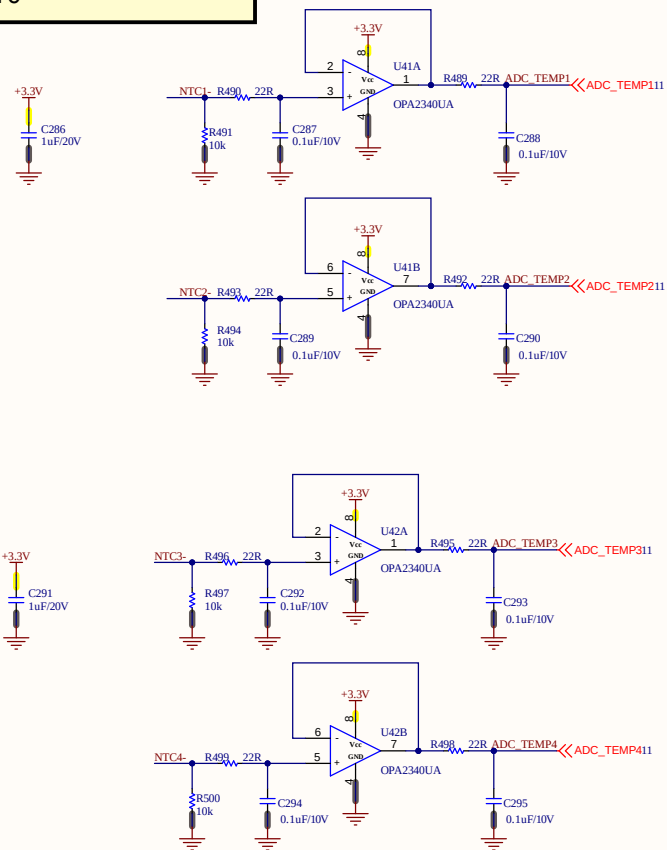
Size: A3

Approved by: Vu Pham

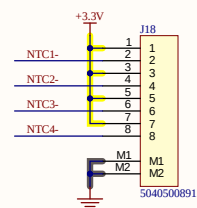
Revision: 1.0.2

Date: 07/07/2026

NTC



CONNECTOR FOR NTC



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: NTC.SchDoc	Sheet 34 of 34
Checked by: Bao Bui	Document No: 34	Size: A3
Approved by: Vu Pham	Revision: 1.0.2	Date: 07/07/2026